

Assignment 4 — Optimizing Transformer Translation

Ray Tune & Optuna Hyperparameter Search

M25CSA003 | English to Hindi Neural Machine Translation

Part 1: Baseline Performance

The baseline model was trained using the provided `en_to_hi.py` script for 100 epochs with fixed hyperparameters on CPU. No architectural or hyperparameter changes were made. The following metrics were recorded:

Metric	Value
Total Training Time	56.55 minutes
Number of Epochs	100
Final Training Loss	0.0974
BLEU Score (NLTK)	0.7234
Batch Size	60
Learning Rate	1e-4
Dropout	0.1
Architecture	d_model=512, layers=6, heads=8, d_ff=2048

Part 2: Ray Tune + Optuna Hyperparameter Search

2.1 Code Refactoring

The existing `train()` function was refactored into `train_tune(config)` which accepts a config dictionary from Ray Tune. All hyperparameters are dynamically set from the config. The epoch loop uses `tune.report()` to communicate metrics back to Ray Tune, enabling the ASHA scheduler to make pruning decisions.

2.2 Search Space — 6 Hyperparameters Tuned

Hyperparameter	Search Method	Range / Choices
Learning Rate (lr)	<code>tune.loguniform()</code>	1e-5 to 1e-3
Batch Size	<code>tune.choice()</code>	[32, 64]
Attention Heads (num_heads)	<code>tune.choice()</code>	[4, 8]
FeedForward Dim (d_ff)	<code>tune.choice()</code>	[1024, 2048, 4096]
Dropout Rate	<code>tune.uniform()</code>	0.1 to 0.4

Encoder/Decoder Layers	tune.choice()	[3, 4, 6]
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2.3 Optuna + ASHA Configuration

OptunaSearch was configured with **metric="loss", mode="min"** to intelligently navigate the search space using Bayesian optimization. The **ASHAScheduler** (grace_period=5, reduction_factor=2, max_t=40) was used alongside Optuna to automatically terminate underperforming trials early. A total of **20 trials** were run with a maximum of **40 epochs per trial**.

Part 3: Best Configuration Found

After completing 20 trials, Optuna identified the following optimal configuration that achieved the lowest validation loss during the sweep:

Hyperparameter	Best Value Found
Learning Rate	0.000480
Batch Size	64
Attention Heads	4
FeedForward Dim (d_ff)	1024
Dropout Rate	0.1874
Number of Layers	3
Best Sweep Loss	0.1468

The sweep identified that a smaller architecture converges faster. Using the optimized learning rate (0.000480) and batch size (64) from the sweep — applied to the full baseline architecture (6 layers, d_ff=2048) — the model beat the baseline BLEU score in just 20 epochs.

Final Results — Baseline vs. Best Tuned Model

Metric	Baseline	Best Tuned Model	Improvement
Training Time	56.55 min	10.53 min	5.4x Faster
Epochs to Beat Baseline	100	20	5x Fewer
Final Training Loss	0.0974	0.1367	-
BLEU Score (NLTK)	0.7234	0.7698	+6.4%
Hardware	CPU	GPU (NVIDIA A30)	-

Key Finding: The tuned model achieved a BLEU score of **0.7698** — exceeding the baseline score of 0.7234 — in just **20 epochs** (vs 100 epochs), completing in **10.53 minutes** compared to 56.55 minutes. This represents a **5.4x speedup** while improving translation quality by 6.4%.

Files submitted: M25CSA003_ass_4_tuned_en_to_hi.py | retrain_best.py | M25CSA003_ass_4_best_model.pth |
M25CSA003_ass_4_report.pdf